



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2014 HSC Course Assessment Task 4

Friday August 15, 2014

General instructions

- Working time – 55 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 2)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M3A – Mr Zuber

☐ 12M4A – Ms Ziaziaris

☐ 12M3B – Mr Berry

☐ 12M4B – Mr Lam

☐ 12M3C – Mr Lowe

☐ 12M4C – Mr Ireland

Marker's use only.

QUESTION	2-1	5	6	7	Total	%
MARKS	$\frac{4}{4}$	$\frac{12}{12}$	$\frac{10}{10}$	$\frac{11}{11}$	$\frac{37}{37}$	

Section I

4 marks

Attempt Question 2 to 1

Mark your answers on the answer grid provided.

Questions	Marks
1. What is the sum of the coefficients in the expansion of $(1 + x)^n$?	1
(A) 0	(C) 2^n
(B) 1	(D) None of the above
2. Which of the following is an equation for simple harmonic motion?	1
(A) $x = a \sin(nt + \alpha)$	(C) $x = a \cos nt + b \sin nt$
(B) $x = a \cos nt$	(D) All of the above
3. What is the coefficient of x^3 in the expansion of $(1 - x)^6$?	1
(A) 20	(C) -20
(B) 15	(D) None of the above
4. Which of the following are defining conditions for projectile motion?	1
(A) $\ddot{x} = -g, \ddot{y} = 0$	(C) $\ddot{x} = 0, \ddot{y} = 0$
(B) $\ddot{x} = 0, \ddot{y} = -g$	(D) $\ddot{x} = -g, \ddot{y} = -g$

Answer grid for Questions 2 - 1

1 – (A) (B) (C) (D)
 2 – (A) (B) (C) (D)
 3 – (A) (B) (C) (D)
 4 – (A) (B) (C) (D)

Examination continues overleaf...

Section II

33 marks

Attempt Questions 5 to 7

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 5 (12 Marks)

Commence a NEW page.

Marks

- (a) An object is moving with displacement x , velocity v and time t . Show that **3**

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \ddot{x}$$

(All working must be shown to obtain full marks)

- (b) The acceleration of a particle moving in a straight line is given by $\ddot{x} = -2e^{-x}$, where x is the displacement from the origin. Initially, the particle is at the origin with a velocity of 2 ms^{-1} .
- i. Show that $v = 2e^{-\frac{x}{2}}$. **3**
- ii. Describe the behaviour of the particle's velocity as x continues to increase. **1**
- (c) A particle is moving in simple harmonic motion. At the end points of the motion, the acceleration is $\pm 1 \text{ ms}^{-2}$. When the particle is 3 cm from the centre of motion, the speed is $2\sqrt{2} \text{ cm s}^{-1}$. **5**

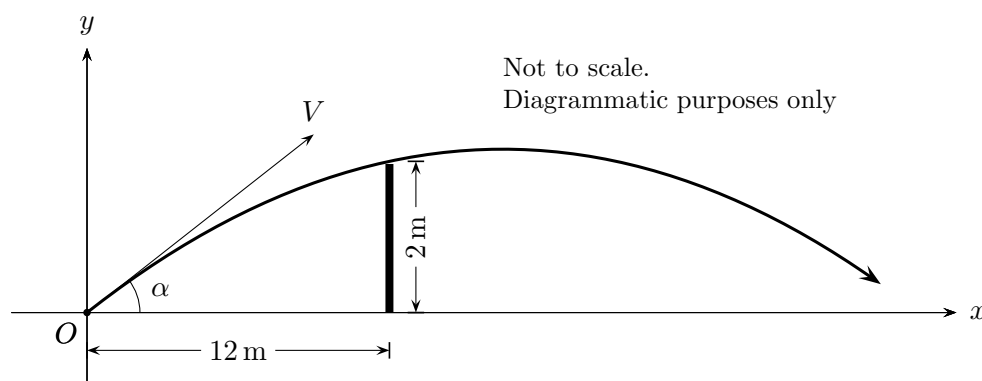
Find the period and amplitude of the motion.

Question 6 (10 Marks)

Commence a NEW page.

Marks

A ball is thrown so that it just clears a wall 2 m high, 12 m from the thrower after 1 s.
Take $g = 10 \text{ ms}^{-2}$.



- (a) Derive the equations of motion for this projectile. 2
- (b) Find the initial projection speed V of the ball, to the nearest metres per second. 2
- (c) Find the angle α at which the ball was thrown, correct to the nearest degree. 2
- (d) Find the maximum distance that the ball could be thrown with the same projection speed, such that it would also clear the wall. 3
- You may use the rounded off value from part (b).
- (e) Copy or trace the diagram above into your working booklet. 1

Show also on the diagram, the altered trajectory if the ball passes through a thin sheet placed above the wall, which reduces its velocity.

Question 7 (11 Marks)

Commence a NEW page.

Marks

- (a) Find the term independent of x in the expansion of 3

$$\left(\frac{x^2}{2} - \frac{3}{x^3}\right)^{10}$$

- (b) Find the greatest coefficient in the expansion of $(3 + 2x)^{12}$. 4
- (c) In the expansion of $(1 + x)(a - bx)^{12}$, the coefficient of x^8 is zero. 4

Find the value of $\frac{a}{b}$ in simplest form.

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Suggested solns

① D ② A ③ B ④ C each

⑤ a) LHS = $\frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \times \frac{dv}{dx} \quad \text{--- ①}$$

$$= v \frac{dv}{dx} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \quad \text{--- ①}$$

$$= \frac{dx}{dt} \times \frac{dv}{dx}$$

$$= \frac{dv}{dt} = \ddot{x} = \text{RHS} \quad \text{--- ①}$$

b) $\ddot{x} = -2e^{-x}$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -2e^{-x} \quad \text{--- ①}$$

$$\frac{1}{2} v^2 = 2e^{-x} + C$$

$$v = 2 \quad x = 0$$

$$2 = 2 + C$$

$$C = 0$$

$$\therefore \frac{1}{2} v^2 = 2e^{-x}$$

$$v^2 = 4e^{-x}$$

$$v = \pm 2e^{-\frac{x}{2}} \quad (\text{initially } v > 0) \quad \text{--- ①}$$

$$v = 2e^{-\frac{x}{2}}$$

$$\therefore \text{As } x \rightarrow \infty \quad v \rightarrow 0 \quad \text{--- ①}$$

$$(c) \quad \ddot{x} = -n^2 x$$

$$|-n^2 a| = 1$$

$$n^2 a = 1 \quad a > 0$$

$$a = \frac{1}{n^2}$$

$$v^2 = n^2 (a^2 - x^2)$$

$$8 = n^2 (a^2 - 9)$$

—— ①

$$8 = \frac{1}{a} (a^2 - 9)$$

$$8a = a^2 - 9$$

—— ①

$$a^2 - 8a - 9 = 0$$

$$(a - 9)(a + 1) = 0$$

$$a = -1 \text{ or } a = 9$$

but $a > 0$ } —— ①

$$\therefore a = 9$$

$$\text{so } n^2 = \frac{1}{9}$$

$$n = \frac{1}{3}$$

—— ①

$$P = \frac{2\pi}{n} = \frac{2\pi}{\frac{1}{3}} = 6\pi \quad \text{—— ①}$$

6

a)

$$\ddot{x} = 0$$

$$\ddot{y} = -10$$

$$\dot{x} = c_1$$

$$\dot{y} = -10t + c_2$$

$$t=0 \quad \dot{x} = V \cos \theta \quad \dot{y} = V \sin \theta$$

$$c_1 = V \cos \theta$$

$$c_2 = V \sin \theta$$

$$\dot{x} = V \cos \theta$$

$$\dot{y} = -10t + V \sin \theta$$

$$x = Vt \cos \theta + c_3$$

$$y = -5t^2 + Vt \sin \theta + c_4$$

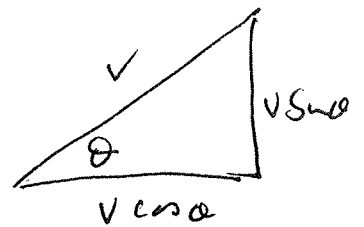
$$c_4 = 0$$

$$y = -5t^2 + Vt \sin \theta$$

$$t=0 \quad x=0 \quad y=0$$

$$c_3 = 0$$

$$x = Vt \cos \theta$$



2

b)

$$t=1 \quad 12 = V \cos \theta \quad 7 = V \sin \theta$$

$$\tan \theta = \frac{7}{12}$$

$$\theta = 30^\circ 15'$$

$$\therefore V = \frac{12}{\cos 30^\circ 15'} = 14 \text{ m/s}$$

c)

$$\theta = 30^\circ 15' = 30^\circ \text{ (nearest degree)}$$

d)

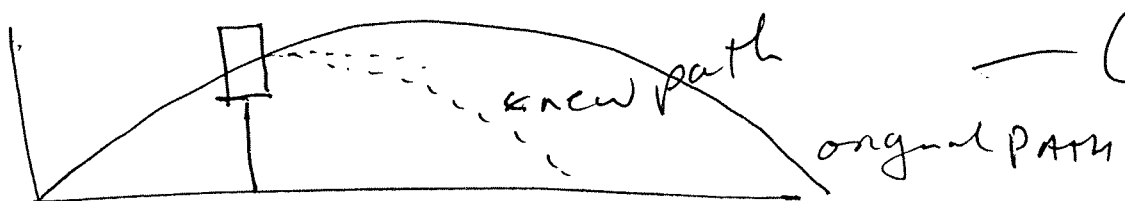
$$y=0 \quad x = \frac{V^2 \sin \theta \cos \theta}{5}$$

$$\theta = 45^\circ \quad V = 14$$

$$x = \frac{14^2 \sin 45^\circ \cos 45^\circ}{5}$$

$$= 19.6 \text{ m}$$

e)



$$\textcircled{7} a \left(\frac{x^2}{2} - \frac{3}{x^3} \right)^{10} T_{rm} = {}^{10}C_r \left(\frac{x^2}{2} \right)^{10-r} \left(\frac{-3}{x^3} \right)^r$$

independent term when $x^{20-2r} \times x^{-3r} = x^0$ — (1)

$$20 - 5r = 0 \quad \text{--- (1)}$$

$$r = 4.$$

$$\text{14 } T_5 = {}^{10}C_4 \left(\frac{x^2}{2} \right)^6 \left(\frac{-3}{x^3} \right)^4$$

$$\text{coeff} = {}^{10}C_4 \left(\frac{1}{2} \right)^6 (-3)^4 \left[\frac{8505}{32} \right] \text{--- (1)}$$

$$\text{4) } \frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \times \frac{b}{a} \quad \text{--- (1)}$$

$$= \frac{13-r}{r} \times \frac{2}{3}$$

$$= \frac{26-2r}{3r}$$

$$\text{--- (1)}$$

$$\frac{T_{r+1}}{T_r} > 1 \text{ when}$$

$$\frac{26-2r}{3r} > 1$$

$$26-2r > 3r$$

$$5r \leq 26$$

$$r \leq 5.2$$

$$r = 5$$

$$\text{--- (1)}$$

$$\therefore \text{Greatest coefficient } T_6 = {}^{12}C_5 (3)^7 (2)^5 \text{--- (1)}$$

$$c) (1+x)/(a-bx)^{12}$$

coeff of x^8 is

$${}^{12}C_8 (-b)^8 (a^4) + {}^{12}C_7 (-b)^7 (a^5) = 0 \quad - (2)$$

$${}^{12}C_8 a^4 b^8 = {}^{12}C_7 a^5 b^7$$

$$\frac{a}{b} = \frac{{}^{12}C_8}{{}^{12}C_7} = \frac{\frac{12!}{4!8!}}{\frac{12!}{5!7!}} \quad - (1)$$

$$= \frac{5}{8} \quad - (1)$$